Hatixhe Merkaj

ID: 98515073794 | **Date of birth:** 09/02/2005 | **Place of birth:** Vlore, Albania | **Nationality:** Albanian | **Phone:** (+90) 5016702005 (Mobile) | **Email address:** hatixhemerkaj05@gmail.com | **Address:** Türkiye (Home)

ABOUT MYSELF

As a 3rd-year Electronics and Communication Engineering student at Istanbul Technical University, I have had the opportunity to learn from leading professors in the field. This experience has allowed me to build a strong theoretical foundation while also applying my classroom knowledge under professional supervision. My work at the ITU Electro-Optical Devices Laboratory has been especially valuable, giving me hands-on experience with the applied aspects of electronics and deepening my understanding of engineering research and practical design. Also joining ITU Hyperbee project team has expanded my technical understanding and also has given me great opportunities to meet people who are professionals in the engineering and research field.Outside academics, I have worked as an English to Albanian and German to Albanian translator, and have also assisted in business and accounting tasks, which strengthened my communication, organization, and teamwork skills.

WORK EXPERIENCE

18/08/2025 - 06/09/2025 - ISTANBUL, TÜRKIYE

ELECTRONICS AND OPTICAL ENGINEERING INTERN PROF. DR. ONUR FERHANOGLU, ITU ELECTRO-OPTICAL DEVICES LABORATORY

-Optical Simulation

-Programming in C

-Researching

09/10/2025 - CURRENT - ISTANBUL, TÜRKIYE

CONTROL AND LEVITATION ENGINEER ITU HYPERBEE PROJECT TEAM

To be able to implement the control system a wide range of literature about Hybrid-Electromagnetic-Suspension Systems was covered. this gave me the oportunity to come out with a new control strategy which merged two methods: cascaded control and gain-schedulling. To succsessfuly utilize that I:

Designed and stabilized a cascaded control architecture (Inner PI Current Loop, Outer Scheduled PD Position Loop) to levitate a 30kg unstable non-linear magnetic system.

Implemented Model-Based Systems Engineering (MBSE) by mathematically deriving local linearizations from high-resolution ANSYS 2D Force/Flux information tables and implementing the final non-linear model into Simulink.

Applied advanced linear control theory , utilizing Root Locus, Bode, and Nyquist criteria to guarantee stability, while strictly enforcing a 10x bandwidth separation rule to prevent subsystem interference.

Developed dynamic gain-scheduling algorithms (1D/2D Lookup Tables) with feedforward voltage and current compensation to handle massive exponential shifts in gravitational and magnetic forces.

Engineered for real-world hardware limits, utilizing Anti-Windup Clamping and retuning system bandwidths to successfully catch the pod without exceeding a strict 50V power supply saturation limit.

01/07/2024 - 01/09/2024 - VLORE, ALBANIA

ENGLISH TO ALBANIAN AND GERMAN TO ALBANIAN TRANSLATOR FOR LEGAL AND BUSINESS MATTERSFAMILY BUSINESS

-Translating

LANGUAGE SKILLS

Mother tongue(s): **ALBANIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
GERMAN	B1	B1	A2	A2	B1
TURKISH	B2	B2	B1	B2	B1
FRENCH	A2	A2	A2	A2	A2

CERTIFICATIONS

17/03/2025 Udemy
Introduction to Optical Eye Modeling with Zemax

Project Based -

Links https://www.udemy.com/course/introduction-to-optical-eye-modeling-with-zemax/ | https://udemy-certificate.s3.amazonaws.com/pdf/UC-f6cb5f22-f7c9-4f6b-9275-2f006502752b.pdf

26/12/2025 Mathworks Academy
Simulink fundamentals Mathworks course

Work based -

Link https://matlabacademy.mathworks.com/details/simulink-fundamentals/slbe

03/03/2025 Udemy
Zemax Sequential Basics: How to Build Lens Systems

Project Based -

Link https://udemy-certificate.s3.amazonaws.com/pdf/UC-ad5b8708-878d-49b5-b495-e92ba98c48c3.pdf

PROJECTS

10/10/2025 - CURRENT

ITU Hyperbee MAthematically modelling the levitation system and designing the PID controller

18/08/2025 - CURRENT

Laser filter simulation Designing, simulating and optimizing a laser system to achieve the nedded parameters in order for the beam to be later used in eye surgical procedures

09/02/2026 - 09/04/2026

Designed and simulated a Wideband Intermediate Frequency (IF) receiver chain for a superheterodyne architecture using the AWR Design Environment. In my recent term project, I successfully completed the system-oriented design and verification of a Wideband Intermediate Frequency (IF) receiver chain using the AWR Design Environment. Tasked with targeting a 70 MHz center frequency, I designed a complete RF-to-IF translation system that included a mixer, local oscillator, two-stage IF amplifier, and a custom-calculated N=3 Butterworth LC band-pass filter. By running rigorous S-parameter and Harmonic Balance simulations, I was able to verify the system's ability to isolate a 5 MHz channel while maintaining approximately 31 dB of total gain. This project required me to move beyond ideal behavioral models to test true system performance, giving me practical experience in diagnosing signal leakage and troubleshooting impedance loading

HOBBIES AND INTERESTS

Literature Classic literature centering authors like Dostoyevski, Stephan Zweig, Reiner Maria Rilke, reading and discussing with my book club

Horse Riding